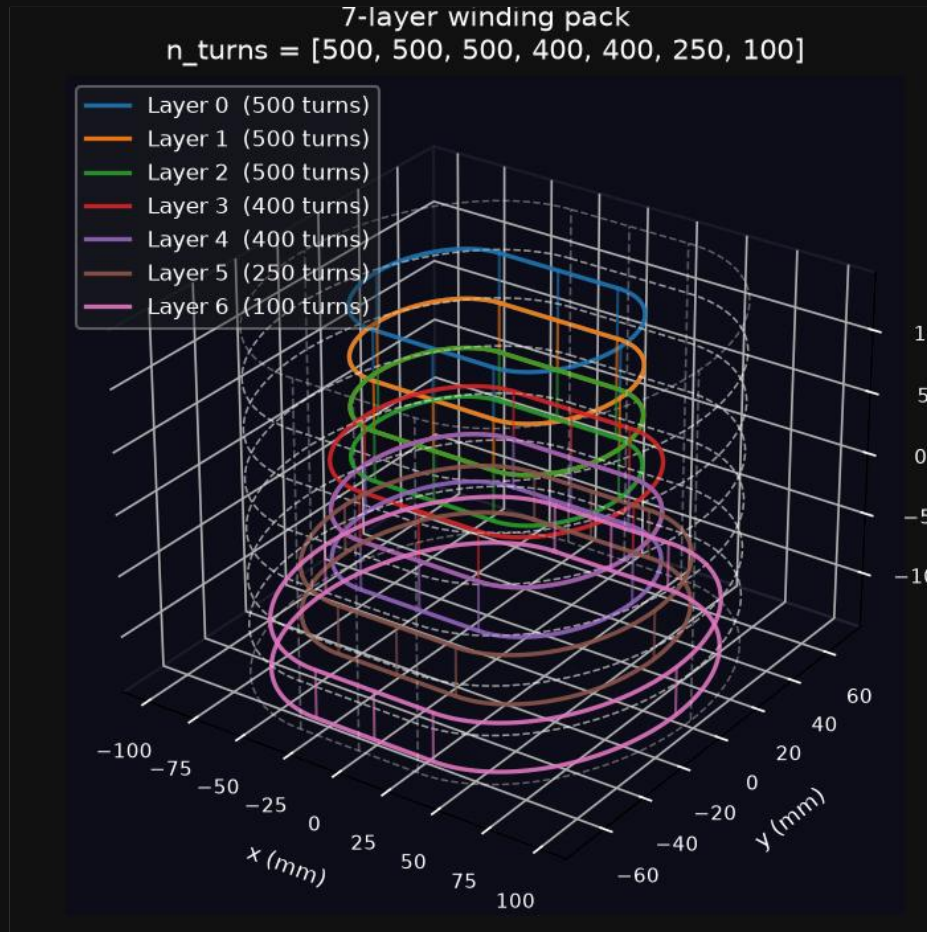

T–A Homogenized FEM Simulation for Shape Optimization of Helmholtz Racetrack Coils

FEniCSx simulation including screening-current, quench margin, and mechanical stress in a two-coil racetrack winding pack

The Coil and the Design Target



7-layer REBCO winding pack, one of two racetrack coils (eighth-symmetry FEM domain)

- ▶ Two racetrack coils (mirror pair) where the turns of each pancake is varied
- ▶ Coils have same outer diameter (less turns = larger inner diameter)
- ▶ Tape: 4 mm wide, 1 μm SC layer, 75 μm radial pitch per turn
- ▶ Design targets:
 - Maximize field in the bore (without quenching – safety factor)
 - < 1% field uniformity over the bore target box
 - No Mechanical Failure
 - Minimize length of tape needed

We simulate only 1/8 of the coil and use symmetry to predict the rest of the system

How the Simulation Works: the T–A Formulation

► Homogenized model: the REBCO tape is treated as a thin anisotropic sheet carrying a scalar current potential T (A/m), coupled to the vector potential A of Ampere's law

► Definition of T in terms of J : $J_{SC} = \nabla T \times \hat{n}$

► Weak Form of T-A Relation (in the coils):

$$\int_{\Omega_T} \rho_{SC} (\nabla T \times \hat{n}) \cdot (\nabla \varphi \times \hat{n}) dx = -\frac{1}{\Delta t} \int_{\Omega_T} (\nabla \times \mathbf{A}_h) \cdot (\varphi \cdot \hat{n}) dx \quad \forall \varphi \in V_T$$

► Boundary conditions: $T_{top} = -I / (2\delta_{SC})$, $T_{bot} = +I / (2\delta_{SC})$ at tape edges

► A-equation source term: $J_s = (\delta_{SC} / \Lambda) \cdot J_{SC}$ (where Λ is the tape pitch)

► E–J power law: $E = E_c (J / J_c)^n$, resistivity $\rho_{SC} = (E_c / J_c) \cdot (J / J_c)^{n-1}$

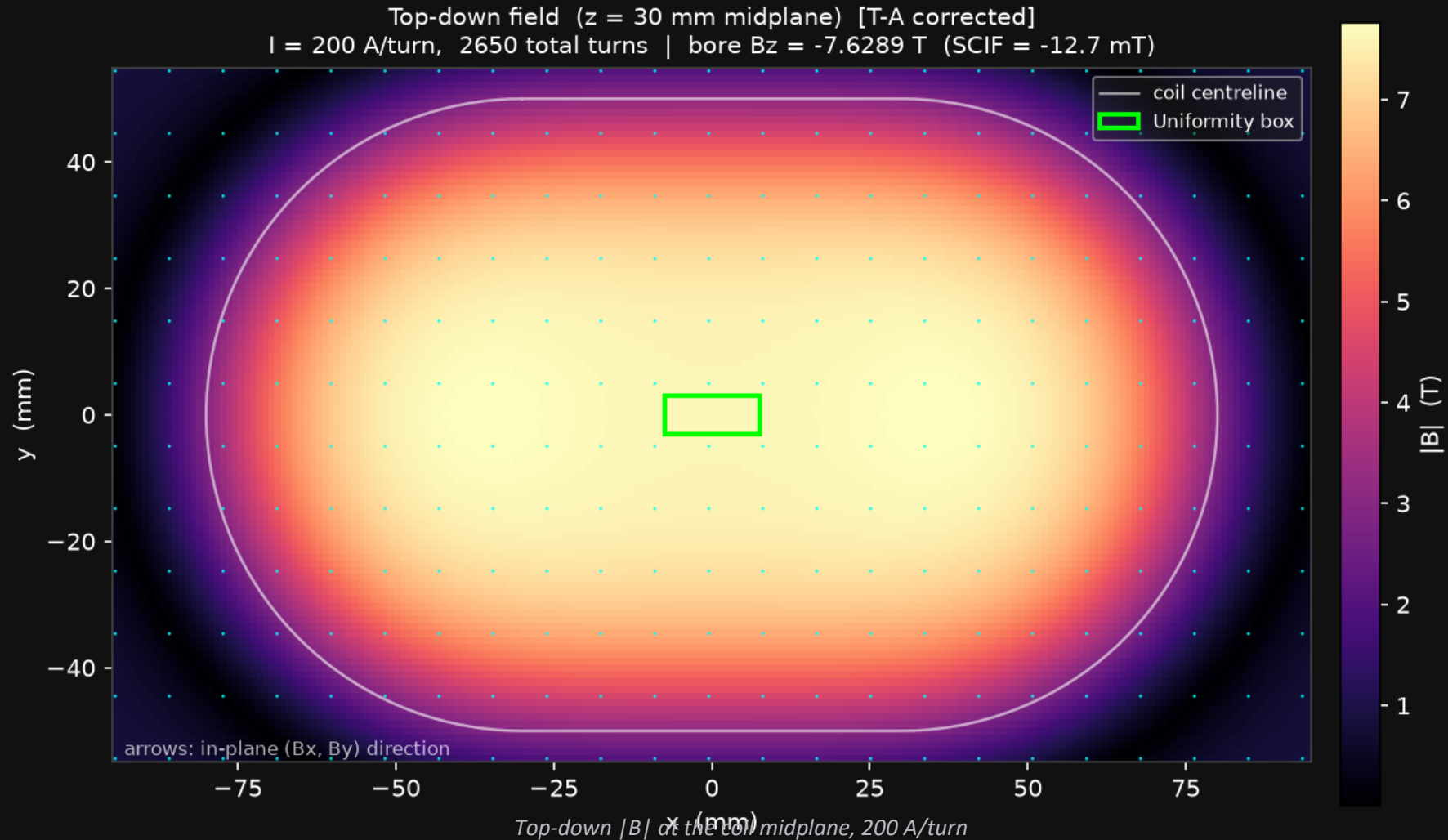
► Method from: *Vargas-Llanos et al. 2022, Supercond. Sci. Technol. 35, 124001*

Picard iteration (per current step)

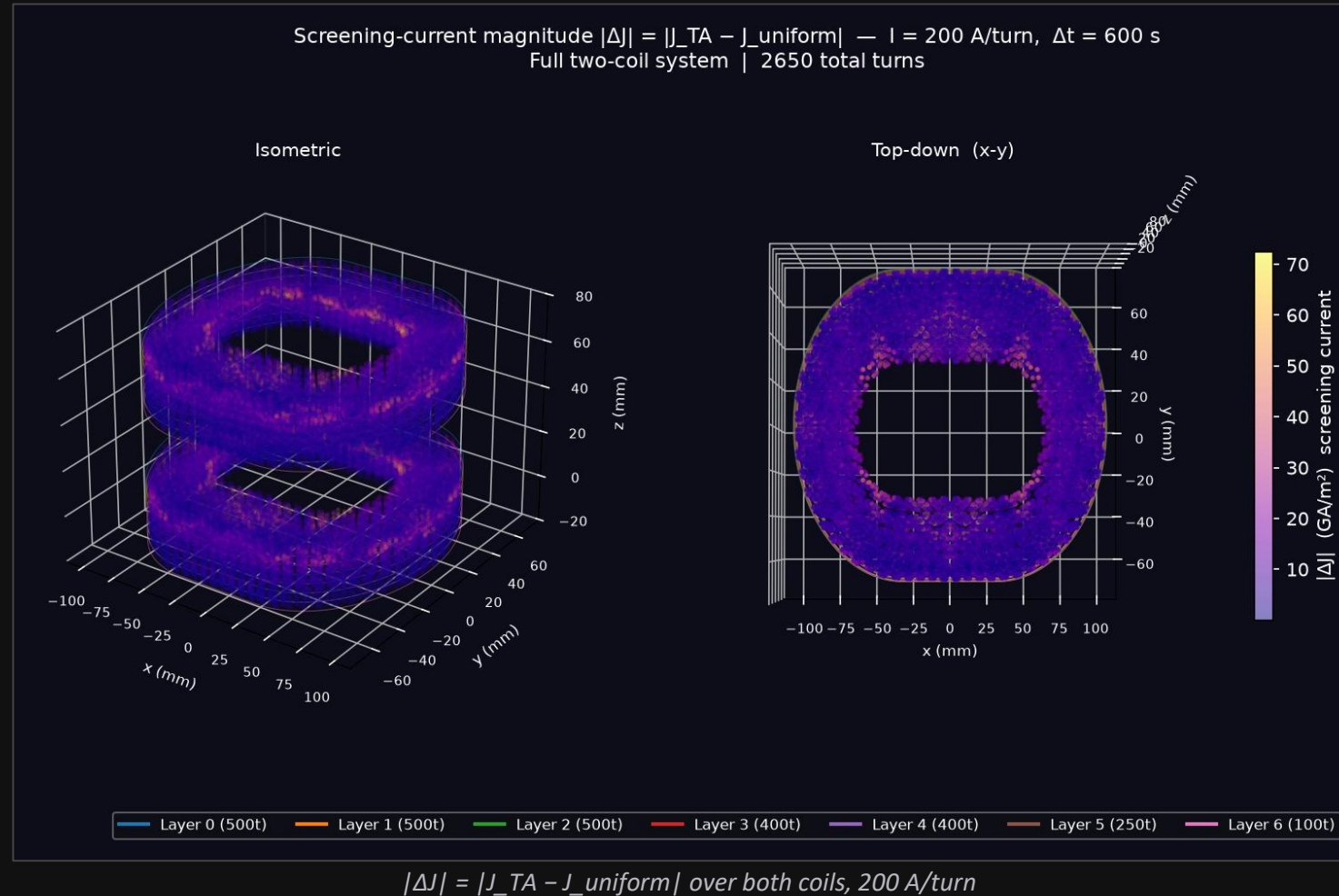
1. Solve linear T-equation with frozen ρ_{SC}
2. Get new T : $(1-\alpha)T_{old} + \alpha T_{new} \rightarrow T$
3. Get J from T , do this once for each z-layer (constant radially)
4. Solve A-equation $\rightarrow$ get new B field
5. Update $\rho_{SC}(J, B)$ from the power law
6. Check convergence

Two-phase relaxation ($\alpha = 0.30 \rightarrow 0.15$); converges in ~ 30 – 80 iterations, ~ 0.2 s/iteration.

Example results of one simulation (top-down view)



Screening Currents: the T–A Correction to a Uniform-J Model

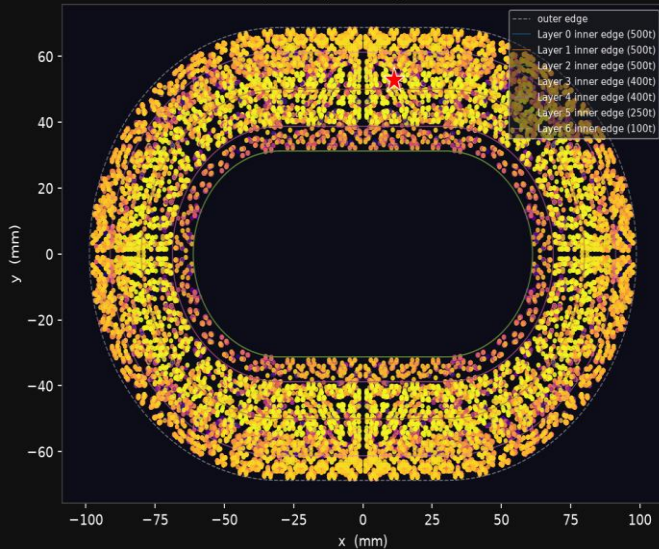


- SCIF is calculated as the absolute value of difference between the field as if we used uniform current and the field from the real current

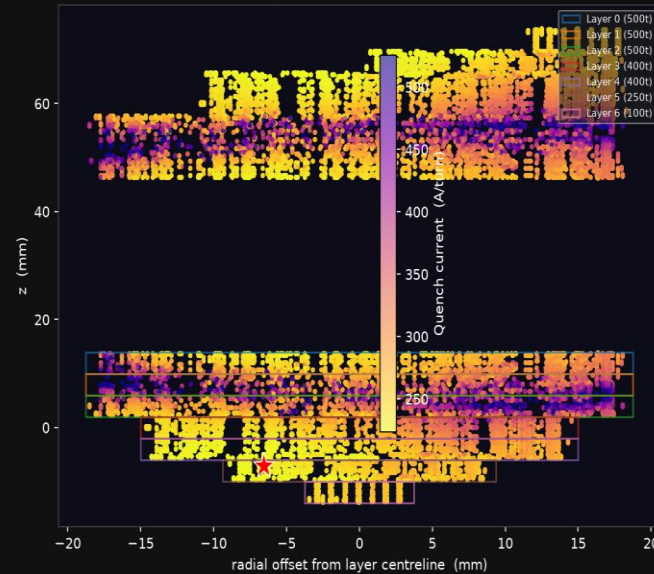
Constraints: Quench Margin and Mechanical Stress

2-D quench current map — $n_turns = [500, 500, 500, 400, 400, 250, 100]$ (2650 total turns)
★ = weakest point (220 A/turn)

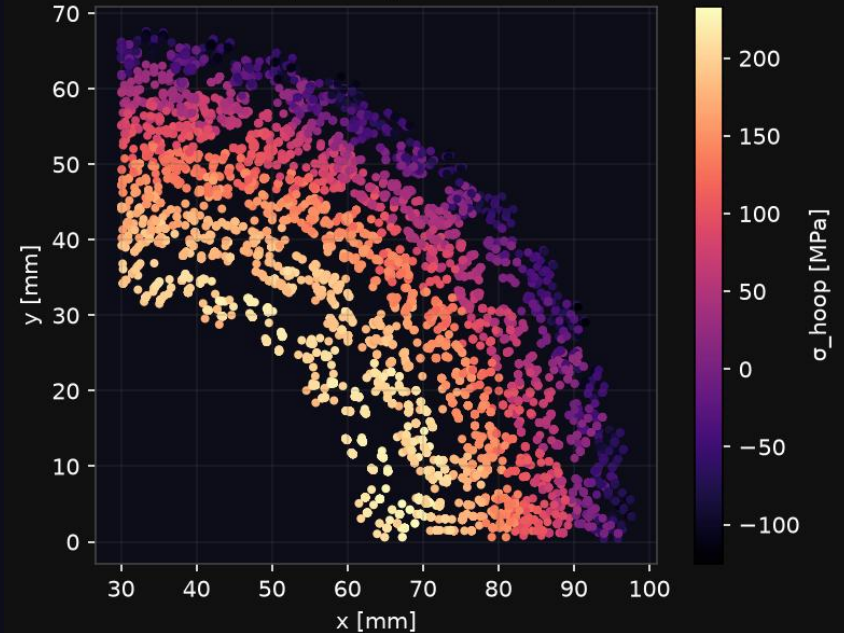
Top-down quench map (all cells)



Radial x axial quench map
(shows which layer and which radial position is weakest)



Cap hoop stress (max 233 MPa)



Cap hoop stress at 200 A

Per-cell quench current map — weakest cell quenches at 220 A/turn

- ▶ Maximum current is limited by where the system quenches. Example of one of these quench locations is shown above
- ▶ Maximum allowable stress is also something that should be considered in the design of one of these coils

What We Are Optimizing

- ▶ Objective: Highest mean field at midplane minimizing the tape length, subject to the following constraints:
 - ≤1% SCIF-corrected uniformity constraint
 - within user-set mechanical stress and strain limits
- ▶ Free parameters:
 - inner/outer radius (a, b)
 - number of turns for each layer (num_turns[i])
 - number of layers (len(num_turns))
- ▶ First do full magnetic field calculation and do full stress only for final candidates
- ▶ Next step: Implement optimization to find geometry ideal candidates